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VLSI Lab 2

Schematic:

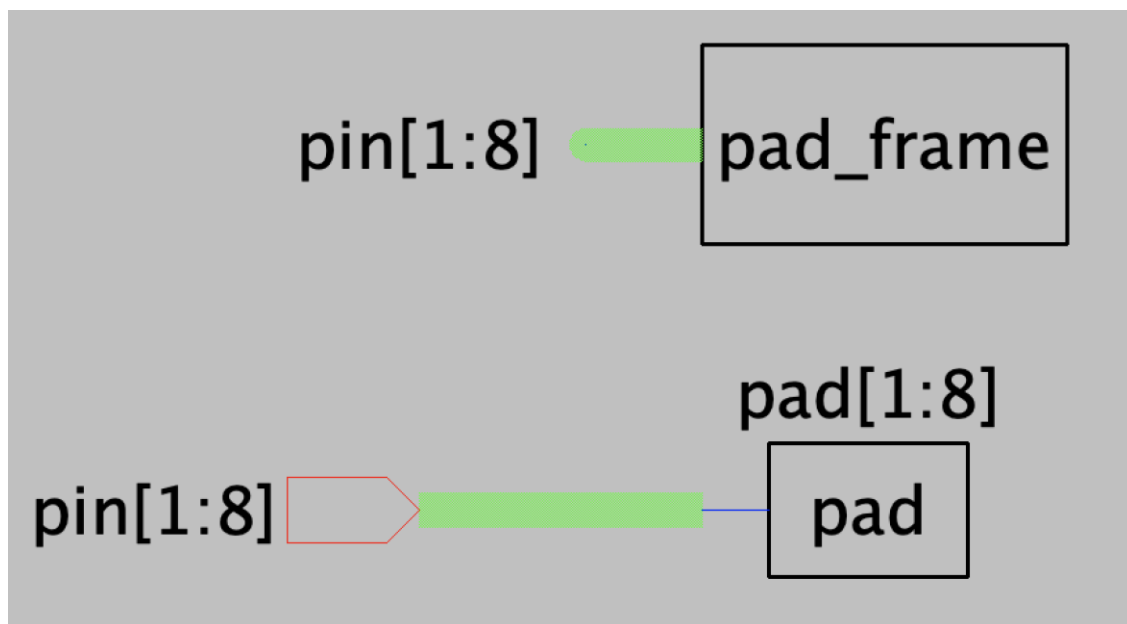


Figure 1: Pad Frame Schematic

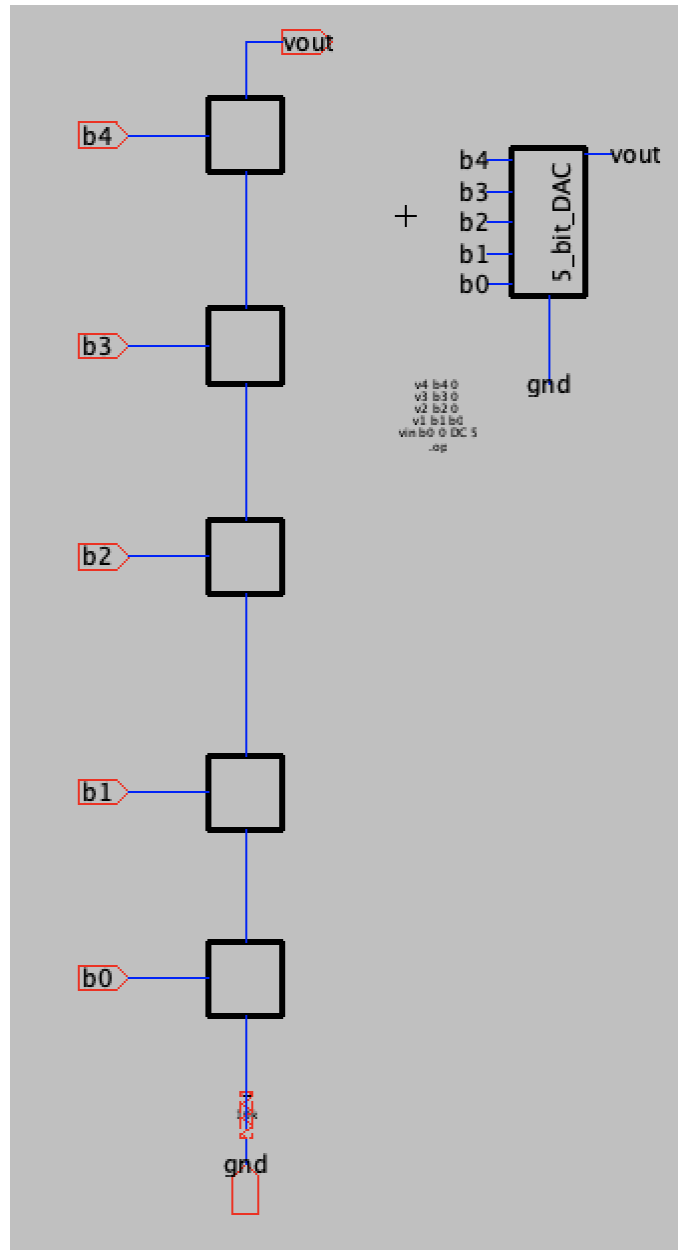


Figure 2: DAC Schematic

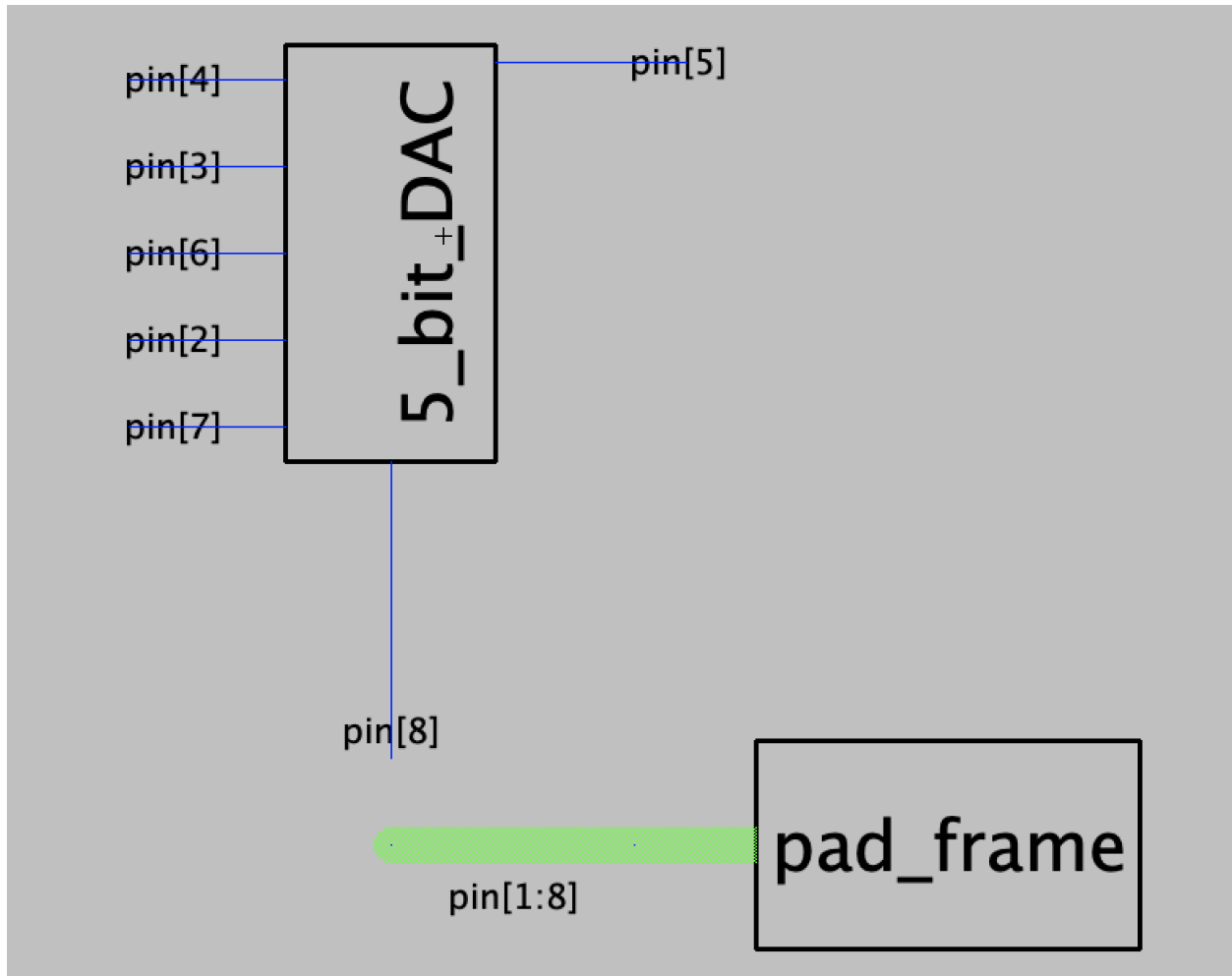


Figure 3: Pad Frame with DAC Schematic

Since the DAC is a 5-bit DAC, it requires 5 input pins, 1 ground pin, and 1 output pin. This means that the DAC requires 7 total pad connections. The pad frame must be a square, so the number of pins must be a multiple of 4. Thus, the minimum number of pins in the pad frame for this design is 8, since 8 is the lowest multiple of 4 which is greater than 7. This leaves one empty pin (pin 1).

Layout:

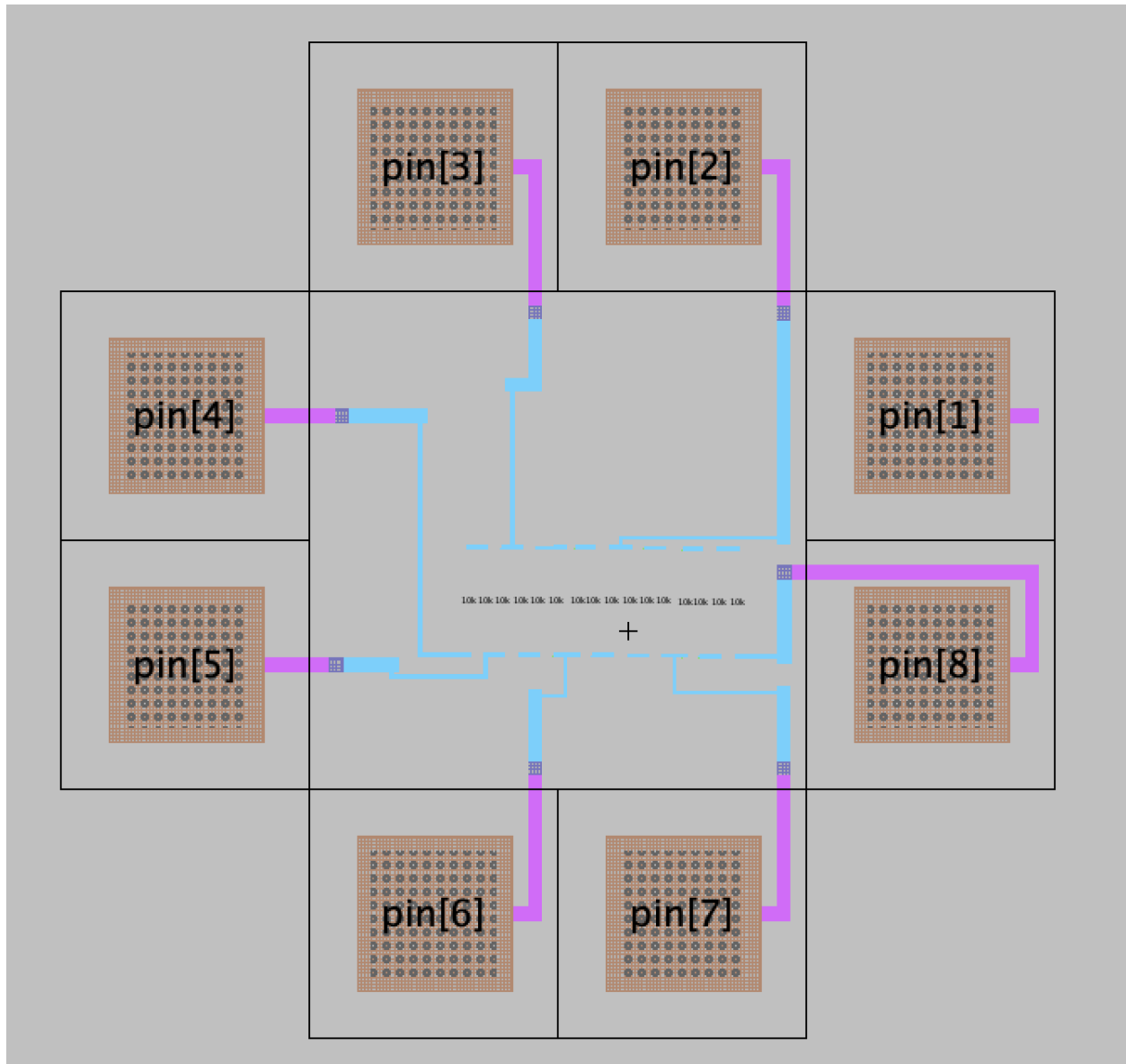


Figure 4: Design Layout

Figure 4 shows the DAC with each pin connected to the appropriate pad within an 8-pin pad frame. Each pin is connected so that no wires cross. The pads and the DAC are on different layers, so vias were used to connect the Metal 1 layer and the Metal 2 layer.